

Designing and Analyzing ChatGPT-based Different Interaction Protocols: a Psychologist Point of View

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Abstract—We present a preliminary work on a tool to improve Human-Robot Interaction (HRI). In this perspective, we explored, from a psychological point of view, whether ChatGPT could be a helpful instrument in adapting to different communication styles. We asked ChatGPT to address three people with varying attitudes of communication: aggressive, distrustful, and pessimistic. We hypothesized different scenarios, including helping the person follow a treatment plan, encouraging sustainable behaviors, and promoting physical activity. This preliminary study aims to estimate the ability of ChatGPT to adapt to different communication styles and, therefore, the possibility of being used for future work in a more complex system that interacts with the user to make HRI more effective.

Index Terms—ChatGPT; Human-Robot Interaction; Human-Computer Interaction; Chatbot; Communication styles;

I. INTRODUCTION

Effective interaction with a person with a specific communication style is a fundamental skill in everyday life, even in social networks [1]. A person's communication style can be defined as their behaviors during interpersonal interaction, including verbal and non-verbal communication. Adapting to the other person's communicative style or at least taking it into account during a conversation allows one to connect with the other person. Even more so, when simulated in robots, this skill can be a fundamental key for an effective and fluent HRI [2]. Once the other person's communicative style is identified, it is crucial to use words and phrases that align with the other person's emotional state and communicative style. In this regard, the contribution of ChatGPT could be essential since, after providing precise prompts representing directives, this tool allows for the rapid generation of words and phrases in line with the identified communicative style with great variety. The prompts representing directives must be provided considering the various communicative styles and understanding their meaning: for example, if a person is angry and has an aggressive communicative style, the phrases and words generated must fall within an "aggressive protocol". In our study, these protocols based on the various communicative styles are drafted, considering psychological knowledge.

An interesting study on ChatGPT and ToM is that one of Marchetti et al (2023) [3]. In their study, the authors explore ChatGPT's ability to understand ToM, noting certain abilities and limitations in its behaviour with respect to understanding

the mental states of others. In [4] ChatGPT has shown to be a useful tool for constructing ad hoc textual datasets for emotion recognition in specific contexts.

In psychology, there are several contributions in the literature on communication styles and their entanglements with personality traits. Hansford and Hattie (1987) [5] conducted a factor structure analysis and identified five dimensions of communicative style (dominant, expressive, relaxed, animated, attentive) and isolated two second-order factors on the scale: animated-dominant and support-attentive.

Other studies have considered the effect of the cultural component. Among these, one of the most recognized is that of Gudykunst et al. (1996) [6]. The research produced the Scale of Communication Styles (CSS) consisting of eight main dimensions (ambiguous and indirect communication, interpersonal sensitivity, positive perception of silence, inference of significance, use of emotion, dramatization, authenticity, and precision). Prominent among studies on communication styles is the contribution of Reinout De Vries et al. (2009) [7]. The authors used a lexical approach to describe communicative behavior. They first searched for underlying factors in a large group of adjectives and verbs in the communicative sphere, extracting seven elements: precision, reflection, expression, support, emotionality, politeness, and threat. Their research led to the construction of the Communicative Styles Inventory (CSI) [8], which, together with Gudykunst's Communicative Styles Scale (CSS), is currently one of the few tools available for the general measurement of the communicative styles adopted by an individual. In the Italian context, Diotaiuti et al. (2020) [9] have done important work in this regard. The authors addressed the translation and validation of the Italian short version of the Communication Styles Inventory (CSI-B/I). Their work makes it possible to measure three main dimensions: impression manipulateness, emotionality and expressiveness.

Other studies consider communication styles not as personality traits but as emerging under the influence of these traits and the environment [10]. According to Schulz von Thun (2003) [11], these styles constitute a specific way of relating to others. Specifically, they are associated with needs, moods, and intentions that are manifested through words and nonverbal signals. The various styles are not mutually exclusive but

are joined in the person in a combination that expresses the individual's preferred relational mode. In this sense, the five-factor model of personality (McCrae et al., 2000) [12] is currently a point of reference for identifying person-specific core tendencies. The five factors are neuroticism, extroversion, openness to experience, friendliness, and conscientiousness.

The current debate on the definition of communicative style includes both the strictly personological aspects and the aspects of social learning and cultural assimilation that inevitably influence a person's communicative style. This means that a person will develop a communication style that tends to be consistent with their basic personality traits but also with their environment and cultural context.

Different studies show that adapting the robot to the personality profile of the user it interacts with makes it more engaging. In addition, humans prefer robots with a personality similar to their own [13], [14], [15]. Instead, in [16] the authors show that sometimes humans prefer a robot with the opposite personality to their own and that this depends on the context. Context is a crucial element in interactions and directs the course of communication style. This is precisely why it is important not to provide the robot with a preconstituted personality but to adapt the personality style according to the situation. In this sense, in [17] authors propose an autonomous learning process to automatically find and adjust the optimal robot personality. Their approach is based on Reinforcement Learning, which draws its rewards directly from real-time social signals to learn how to dynamically target individual human preferences quickly.

In [18], authors focus on linguistic strategies that characterize the personality traits of extraversion and introversion, analyzing features and proposing a verbal communication model for the two personality types. They are inspired by the study of Gill and Oberlander [19], who in their work identified grammatical properties of the two personality traits. Their approach combines psycho-linguistics and statistical Natural Language Processing techniques to replicate previous findings on the expression of two personality types through language.

Another interesting study was discussed in [20] where the system first estimates the personality traits of the individual interactor through a psycholinguistic analysis of spoken language, then utilizes the natural language generator PERSON-AGE, which attempts to generate verbal language corresponding to the estimated personality traits. In addition, gestures are generated by using BEAT toolkit, which analyzes the generated language relying on rules derived from research into human conversational behavior. Using Nao robot, they show that individuals preferred to interact with a robot with the same personality and that a speech and gesture robot's behavior was more engaging and effective than a robot that interacts only with speech. Also, in the context of assistive robots, an interesting study is that of [21]. The authors provide evidence that people also prefer the same personality in the assistive domain and the effectiveness of robot behavior adaptation to user personality. By evaluating the communication style, these studies provide a response based on the same communication

style.

In this paper, the idea is to explore the possibility of assessing a user's communication style and responding with a communication style that best matches the one exhibited by the user. The style of communication used by our system, therefore, may not even mirror that of one of the users; for example, in the case of an aggressive communication style, the system would use a style aimed at calming the user, as a consequence of an appropriate communication strategy behind it.

The study aims to evaluate the effectiveness of ChatGPT in adapting to different communication styles. Its ability to change communication strategies according to the user, taking only text into account, is evaluated. In future work, we also want to include tone of voice to evaluate the effect of the tool in adapting to different communication styles.

The contexts in which this approach can be helpful are varied. The idea is to use protocols designed for certain communicative styles so they can be used to meet the user's needs best. This is all oriented toward better HRI, using basic psychological knowledge about communicative styles. HRI can often be ineffective because the robot fails to communicate with the user, proposing predefined responses and assessing not "how" the person is talking. Presenting suggestions may not be helpful if the person has a confrontational, distrustful or negative communicative attitude. Equipping the robot with the ability to adapt to that communicative style and try to communicate effectively with that specific user could impact the effectiveness of HRI.

II. METHODOLOGY

In our exploration, we used ChatGPT in some hypothetical scenes (suggesting sustainable behaviors, motivating to physical activity, and following a treatment plan).

We asked ChatGPT to propose different communication styles based on three types of people: aggressive, diffident, and pessimistic. In Fig. 1 we show the sequence of the experiment. In particular, after providing instructions, we asked ChatGPT to use the different protocols for each communication style: each communication style is represented by a different colour. Finally, we requested ChatGPT to explain the strategies it used.

Specifically, we analyzed four situations:

- Turning off the light during the day;
- Shutting off the water when not necessary;
- Doing some physical exercise;
- Taking the medicine prescribed by the doctor.

We show an example of one of the situations described. The interaction with ChatGPT was conducted in Italian language; here it is reported a translation into English for publication purposes.

A. Interaction Example

Prompt: *Imagine the scene: you are at home and you have to tell one person to turn off the light because it is daytime,*

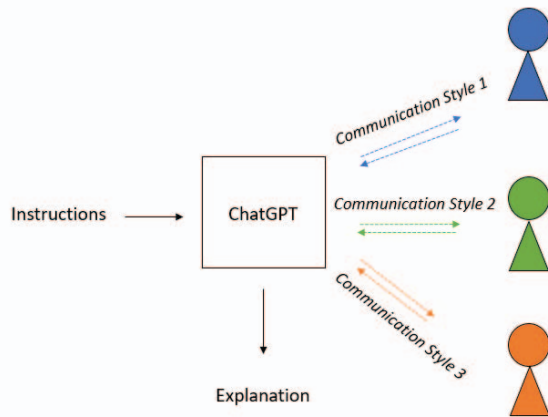


Fig. 1. Sequence of experiment

although he/she keeps it on in order to not open the window and be seen by the neighbors. What would you say to three different people, since one is aggressive, one is diffident and one is pessimistic?

ChatGPT answered:

Here is how I might approach the situation with three different people, each with a distinct personality:

- The aggressive person: "I know keeping it on may be important to you, but what I'm asking is for you to save electricity and help reduce costs. That way, we can strike a balance between your privacy and respect for the environment and finances."
- The diffident person: "I understand that you may have concerns about your privacy, but turning off the light during the day is a common practice that helps save energy. We can find solutions to ensure your privacy in other ways without having to keep the light on."
- The pessimistic person: "I know you might be concerned about your privacy, but remember that turning off the light during the day is a small gesture that can help save on energy costs. It is a simple way to contribute to the environment and save money. We can also explore other options to protect your privacy if you prefer."

In all three cases, the goal is to communicate the need to save electricity in a respectful way and to offer alternatives to address the person's concerns appropriately. The approach varies according to different personalities and concerns.

Prompt: Explain the strategies and reasoning you used for each communication style

ChatGPT answered:

Here are the strategies and reasoning behind each of the responses, taking into account the different personalities of the people involved:

- The aggressive person:

Strategy: I used an approach based on understanding the person's priorities and offering a compromise. **Reasoning:** Aggressive people often focus on their needs and wants. I began by showing understanding for his concern about privacy. Next, I offered a compromise that takes into account his privacy but also encourages energy conservation, a common goal.

- The diffident person:

Strategy: I used an understanding approach, emphasizing common practice and offering alternatives. **Reasoning:** Distrustful people may be reluctant to change their habits. I started by showing empathy for her privacy concerns. Then I pointed out that turning off the light during the day is a common practice, offering an argument based on normality. Finally, I proposed alternatives to address his privacy concern in a different way.

- The pessimistic person:

Strategy: I used an approach based on the importance of saving energy and recognizing concerns. **Reasoning:** Pessimistic people may see mainly the negative aspects. I started by acknowledging his concern about privacy and then explained the importance of energy conservation as a way to reduce costs and environmental impact. I also offered the possibility of exploring alternatives to address his privacy concern.

In all three cases, the goal was to communicate respectfully and offer solutions or alternatives that address people's concerns and promote energy conservation. The strategies varied to accommodate the different personalities and concerns of the people involved.

We proceeded in the flow of communication to see if the difference between personality styles continued.

Prompt: The aggressive person now asks you, "what can be the compromise you are talking about?"

ChatGPT answered:

A possible compromise might be to use lower intensity lights during the day or curtains and drapes that allow you to better manage your privacy without having to keep the light on constantly. In this way, we save electricity while maintaining your privacy. We can also explore additional options together to find the best solution for you.

Prompt: The diffident person now asks you, "what can be the compromise you are talking about?"

ChatGPT answered:

I understand your concerns about privacy. A possible compromise might be to use thicker, opaque curtains or drapes that provide your privacy without having to keep the light on. This way, you can feel safe and save energy. We are open to exploring different solutions to find the one that works best for you and to ensure that you feel comfortable in your home.

Prompt: The pessimistic person now asks you, "what can be the compromise you are talking about?"

ChatGPT answered:

I can understand that you might see the negative aspects of this situation. However, a possible compromise might be to keep the light on only in some rooms and turn it off in those

where you feel uncomfortable. In this way, we reduce energy consumption and allow you to have control over your privacy. In addition, we can explore additional solutions together to ensure that you feel safe without having to keep the light on constantly.

We extended the flow of communication a little more for example with the pessimistic person.

Prompt: *The pessimistic person tells you "keeping the light on conforms me, darkness worsens my mood, but at the same time I would like to have a sustainable behaviors as proposed by you*

ChatGPT answered:

I understand your concerns and you wish to maintain an atmosphere that makes you feel good. Let's find a compromise that can meet both goals. We could explore the use of dimmable lights or lights with warm tones that can help create a cozy and sustainable atmosphere. In this way, you can have lighting that makes you feel good and at the same time reduce energy consumption. What do you think about this option?

III. RESULTS AND THE PSYCHOLOGIST POINT OF VIEW

In the following we shortly discuss the results highlighting the psychologist point of view.

A. Results

As evidenced by the communication flow, ChatGPT shows a good ability to adapt to different people's communication styles in all situations considered as case studies. When asked to explain the strategies used, it provided the requested explanations. Certainly not having direct access to the training of the tool, the analysis conducted is qualitative and inferential, based on the observation of the responses. But despite of this, many of the responses and strategies used can be attributed to interactions based on psychological knowledge.

B. The psychologist point of view

In general, with the aggressive personality, it mainly uses communication strategies that can be referred to as assertive and direct strategies. This is consistent with the main communication modes generally suggested. With an aggressive person, in fact, it is always recommended to break free from the reactive plan and try to communicate calmly but directly about one's needs. ChatGPT used this conversational style to suggest sustainable behavior in this case.

On the other hand, with respect to the diffident communicative attitude it shows that it predominantly selects a communication strategy based on reassurance and trust-building. In fact, diffident people tend not to trust the other person and need to be reassured in order to make decisions.

With the pessimistic communicative attitude, instead, it appears to have adopted a strategy based on emphasising the benefits and therefore the positive aspects of the decisions to be taken. It is reasonable to assume that this approach recognises the tendency of this communication style to focus only on the negative and to feel demoralised. Also going forward in the flow of communication, as we pointed out that the change of

choice would affect the person (turning off the light would affect the mood), ChatGPT showed adaptation by taking the person's mood into consideration.

IV. OTHER USAGE SCENARIOS

The scenarios for using the tool can be different. As mentioned earlier, we have essentially proposed two contexts here: health and sustainable behaviors, assuming use with household assistants. These two contexts are, to date, among the main ones to have a significant impact on the individual and the community. But another scenario in which the use of the tool could be very useful is in public administration. One of the fundamental problems between public administration and the citizen concerns the appropriate communication level. Indeed, in these contexts, frustration levels are sometimes so high that they create misunderstandings and undermine the effectiveness of the operations to be performed. Hypothesizing that AI tools placed in these contexts are equipped with the ability to adapt to the person's communicative style can significantly impact achieving the goal of the interaction. This can extend to all virtual assistants and chatbots that today have among their main problems precisely that of failing to connect with the user, who, after a short time, abandons the interaction with the chatbots to turn to a human operator: think of telephone operators, streaming services, etc.

In addition, it would be interesting to broaden the types of interactions possible, by testing the tool with other communication styles, including, for example, authoritarian. For example, a person with an authoritarian communication style might block interaction from the beginning with a chatbot that is too prescriptive and does not take into account the peculiarities of this communication style.

V. CONCLUSIONS AND PERSPECTIVES

Our preliminary experiments showed that ChatGPT seems to adapt well to different communication styles.

The first advantage, then, is that embedded in a more complex architecture, it can be used to adapt to varying types of users. The creation of a system that 'sorts' the various ChatGPT interactions into different communication styles should be planned. This may potentially provide a module of a larger system that identifies which possible interaction protocol is most appropriate for a specific user.

In this sense, one disadvantage is that only the textual content of ChatGPT cannot be directly linked to different communication styles. For example, the text must be complemented by a tone of voice appropriate to the content of the communication. This limitation, as we said earlier, could be overcome by embedding the tool in a more complex cognitive architecture. Also, clearly, the communicative attitude is built into the interaction and follows a more in-depth flow of communication, so the interaction process must consider other more complex elements.

The purpose of the study, however, was to analyze the potential of ChatGPT in adapting to different communicative styles. On this point, we can affirm that it offers a quite satisfactory mode

of interaction that is well suited to different communication styles. Certainly, one of the main limitations is having performed few trials, since the chatGPT tool can change its results in subsequent executions of the interaction. But, despite of this, in the few, preliminary, experiments conducted it generally maintains the different communicative styles and strategies used during text production.

A. Future Perspectives

In future work, our goal is to repeat the experiment several times, extend the range of cases to take into account, the experiments, and evaluate their outcomes more analytically. Moreover, one goal will be to supplement these qualitative observations with quantitative data.

Furthermore, future work will involve building a more complex system and the component that directs communication based on the user. It will also be interesting to test this system, measuring the impact it has on end users, through the use of questionnaires providing quantitative data and that people can compile to express the quality of the experience. As a matter of fact, a system that can engage with users by adapting its communication style to their attitudes has the potential to create a user enjoyable and efficient experience. This can be especially beneficial for users with different communication needs or preferences.

In our vision, this characteristic offers many advantages. For instance, tailoring communication to preferences and attitudes can enhance the user experience making interactions more pleasant and engaging.

Additionally, when a system is capable of adjusting its communication style to match users preferences, it increases the chances of information comprehension and reduces misunderstandings. Moreover, a system that communicates in a way that resonates with users preferences is more likely to foster interactive relationships. By personalizing the used communication styles, the system can demonstrate an understanding of users needs creating a sense of connection with them. Furthermore, when the system aligns with users preferred communication style, it can facilitate goal achievement by potentially reducing frictions that may arise during the interaction process.

Another advantage of exploiting such a system is its ability to efficiently address issues by understanding users attitudes. For example, some individuals may prefer concise instructions, while others may require explanations. Moreover, the use of such a system can enhance the efficiency of decision support making it possible that users obtain information in a manner that facilitates their decision making process. Besides, it is important to note that users are usually more inclined to place trust in a system that communicates according with their expectations, and this can be valuable for our future research work.

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